Reg. No. | | | | | | | | | | | [18MTEST202]

SECOND SEMESTER M.TECH EXAMINATION, JUNE – 2019

ADVANCED VLSI TECHNOLOGY AND DESIGN

Time: 03 Hours **Maximum Marks:100**

Instructions:

1. Answer **FIVE** full Questions. Question1 and Question2 are compulsory.
2. Missing data may be suitably assumed.

1.	**a.**	List the various methods of design in VLSI systems. Explain any one method in detail.	**08**
	b.	Realize two input NAND and AND gate using the following logics and compare their performances. **i.** Pass Transistor Logic **ii.** Differential cascode voltage switch logic.	**06**
	c.	Draw the schematic and layout of two input NOR gate.	**06**
2.	**a.**	Define setup time, hold time and propagation delay for D flipflop with suitable diagrams.	**06**
	b.	Design negative and positive latches based on multiplexers and TG logic.	**08**
	c.	Explain how to generate a two phase non overlapping clock using D register.	**06**
3.	**a.**	Explain the following VLSI design styles using flow charts **i.** Full Custom design **ii.** FPGA based design	**12**
	b.	What is sheet resistance? Find out the expression of the resistance of rectangular sheet.	**08**

OR

4	**a.**	Explain with neat block diagram phase locked loop (PLL) and its role in clock stability.	**08**
	b.	What is clock skew? Explain with example the implication of clock skew for the performance and functionality of sequential systems.	**12**
5.	**a.**	Derive carry look ahead adder equations and realize 4 bit CLA with a combination of ripple carry adder (hybrid adder).	**10**
	b.	Explain with neat diagram 4 x 4 array multiplier. What are the disadvantages of array multiplier?	**10**

OR

6.	**a.**	Explain the time power reduction techniques used in data path circuits.	**10**
	b.	With relevant diagram explain Barrel shifter with a programmable shift width from 0 to 3. Mention its advantages.	**10**

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7. a. Explain the NOR and NAND based ROM structure with example. **10**

b. Explain the read and write operation of six Transistor SRAM cell with relevant timing diagrams. **10**

OR

8. a. Explain the sources of power dissipation in memories. **10**

b. With a neat diagram explain the architecture of N word memory. **10**